

Small scale fading and Multipath

Small scale fading or fading is used to describe the rapid fluctuations of the amplitude, phases or multipath delays of a radio signal over a short period of time or travel distances, so that large scale path loss effects may be ignored.

Fading is caused by interference between multipath signals at the receiver. This results into wide variations in the amplitude and phases of resultant signal.

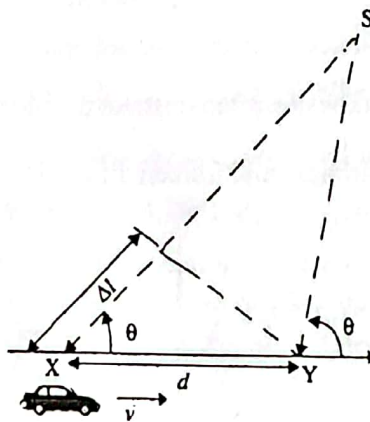
Effects:

1. Rapid changes in signal strength over small travel distance or time interval
2. Random frequency modulation (Doppler Shift)
3. Time dispersion

Factors that influence fading:

1. Multipath propagation
2. Speed of mobile
3. Speed of surrounding object
4. Transmission bandwidth of signal

Doppler Shift



Consider a mobile is moving at a constant velocity v from X to Y separated by a distance d . it receives signal from a remote source S as depicted in the diagram above.

The path difference in this case is given by,

$$\Delta l = d \cos \theta$$

Now,

$$d = v \Delta t$$

$$\Delta l = v \Delta t \cos \theta$$

Phase difference,

$$\Delta \phi = \frac{2\pi}{\lambda} \Delta l$$

$$\Delta \phi = \frac{2\pi}{\lambda} v \Delta t \cos \theta$$

Doppler Shift in frequency is given by,

$$\omega_d = \frac{\Delta \phi}{\Delta t}$$

$$\omega_d = \frac{2\pi}{\lambda} \frac{v \Delta t \cos \theta}{\Delta t}$$

$$\omega_d = \frac{2\pi}{\lambda} v \cos \theta$$

Also,

$$\omega_d = 2\pi f_d$$

$$2\pi f_d = \frac{2\pi}{\lambda} v \cos \theta$$

Hence,

$$f_d = \frac{v \cos \theta}{\lambda}$$

Shift in frequency is thus,

$$f = f_c \pm f_d$$

Parameters of mobile multipath channels

1. Time dispersion parameter
2. Coherence bandwidth
3. Doppler spread and coherence time

Time dispersion parameters

1. Mean excess delay ($\bar{\tau}$)
2. RMS delay spread (σ_τ)
3. Maximum excess delay (X dB)

Mean excess delay:

$$\bar{\tau} = \frac{\sum_k a_k^2 \tau_k}{\sum_k a_k^2} = \frac{\sum_k P(\tau_k) \tau_k}{\sum_k P(\tau_k)}$$

RMS delay spread:

$$\sigma_\tau = \sqrt{\overline{\tau^2} - (\bar{\tau})^2}$$

$$\overline{\tau^2} = \frac{\sum_k a_k^2 \tau_k^2}{\sum_k a_k^2} = \frac{\sum_k P(\tau_k) \tau_k^2}{\sum_k P(\tau_k)}$$

Maximum excess delay is the value on time axis of a power delay profile corresponding to latest arriving path that has strength (dB) above specified threshold.

Coherence Bandwidth

Coherence bandwidth is statistical measure of the range of frequencies over which the channel can be considered flat. Flat channel passes all spectral components with approximately equal gain and linear phase.

It is also defined as the range of frequencies over which two frequency components have strong potential for amplitude correlation.

For 90% correlation, coherence bandwidth is expressed as;

$$B_c \approx \frac{1}{50\sigma_\tau}$$

For 50% correlation, coherence bandwidth is expressed as;

$$B_c \approx \frac{1}{5\sigma_f}$$

Doppler Spread and coherence time

This parameter describes time varying nature of channel, caused by either relative motion between mobile and base station or the movement of objects in the channel.

Doppler Spread (B_D) : Doppler spread is a measure of spectral broadening. It is defined as range of frequencies over which the received Doppler spectrum is essentially non zero.

Coherence time (T_c) : Coherence time characterizes time varying nature of the frequency dispersive channel in time domain.

$$T_c \approx \frac{1}{f_m}$$

Coherence time is statistical measure of the time duration over which the channel impulse response is essentially invariant.

Also can be defined as time duration over which two received signals have a strong potential for amplitude correlation.

For 50% correlation:

$$T_c \approx \frac{9}{16\pi f_m}$$

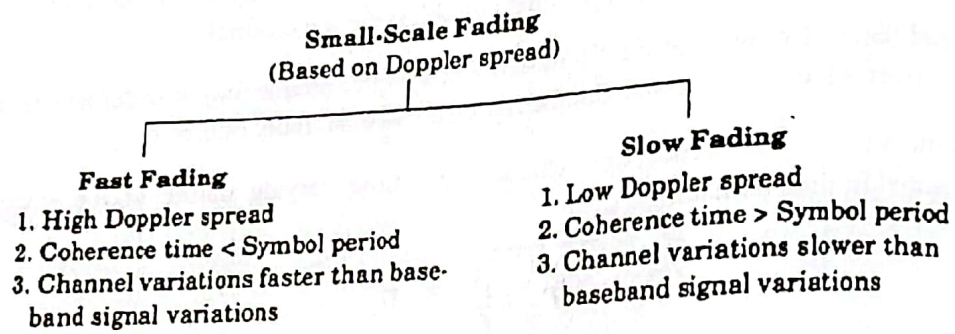
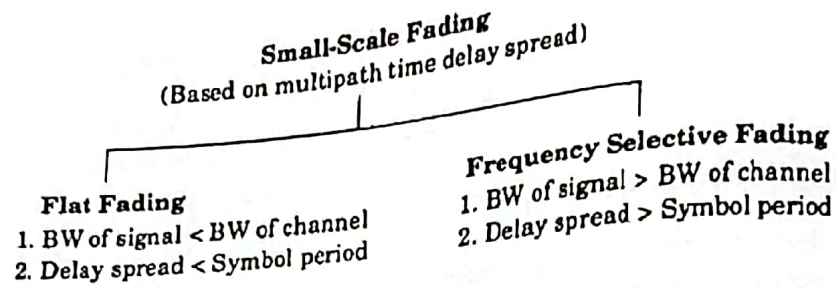
Rule of thumb for coherence time calculation:

$$T_c = \sqrt{\frac{9}{16\pi f_m^2}} = \frac{0.423}{f_m}$$

Types of small scale fading

Depending on the relation between the signal parameters and channel parameters transmitted signals will undergo different types of fading.

1. Multipath delay spread leads to time dispersion and frequency selective fading
2. Doppler spread leads to frequency dispersion and time selective fading



Flat Fading:

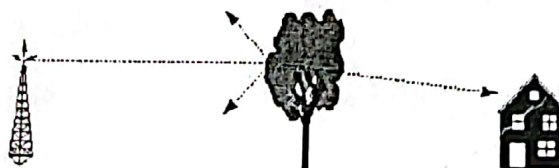
If the mobile radio channel has a constant gain and linear phase response over a bandwidth which is greater than the bandwidth of the transmitted signal then the received signal will undergo flat fading.

A signal undergo flat fading if,

$$B_s \ll B_c$$

and

$$T_s \gg \sigma_t$$



Flat Fading is caused by absorbers between the two antennae and is countered by antenna placement and transmits power level.

Frequency selective fading:

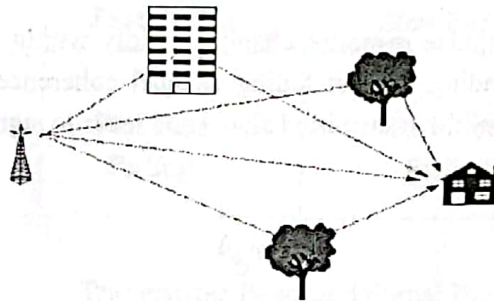
If the channel possesses a constant gain and linear phase response over a bandwidth that is smaller than the bandwidth of transmitted signal then the received signal undergo frequency selective fading.

A signal undergo flat fading if,

$$B_S > B_C$$

And,

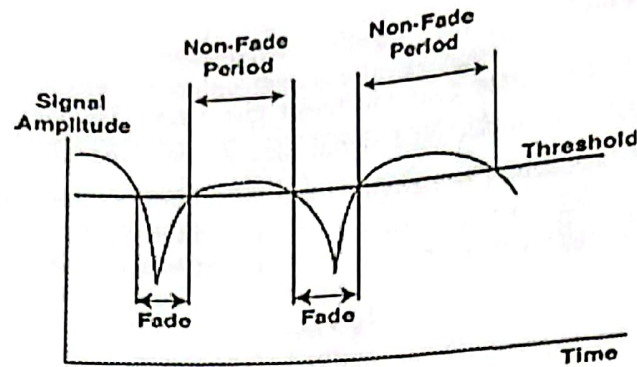
$$T_S < \sigma_\tau$$



Frequency selective fading is caused by reflectors between the transmitter and receiver creating multi-path effects.

Effects of Frequency Selective Fading

- The dips or fades in the response due to reflection cause cancellation of certain frequencies at the Receiver.
- Reflections off near-by objects (e.g. ground, buildings, trees, etc) can lead to multi-path signals of similar signal power to the direct signal.
- This can result in deep nulls in the received signal power due to destructive interference.



When the waves of multi-path signals are out of phase, reduction of the signal strength at the receiver can occur.

Fast fading:

If the mobile radio channel impulse response changes rapidly within the symbol duration the received signal undergo fast fading. In fast fading channel coherence time of the channel is smaller than the symbol period of the transmitted signal and leads to signal distortion.

A signal undergo flat fading if,

$$T_S > T_C$$

And,

$$B_S < B_D$$

Slow fading:

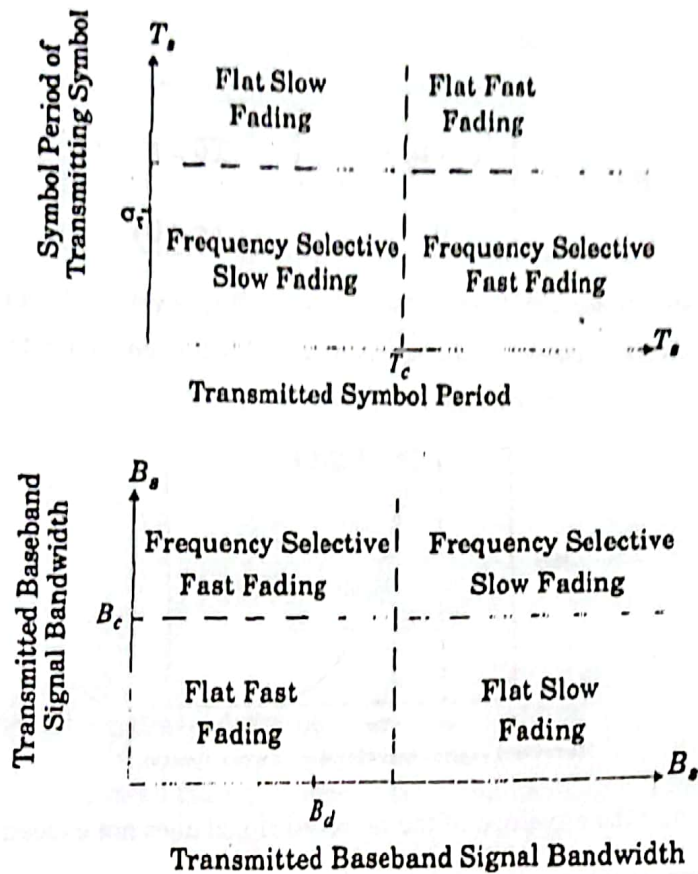
If the mobile radio channel impulse response changes at a rate much slower than the transmitte baseband signal the received signal undergo slow fading. The channel may be assumed to b static over one or several reciprocal bandwidth intervals.

A signal undergo flat fading if,

$$T_S \ll T_C$$

And,

$$B_S \gg B_D$$

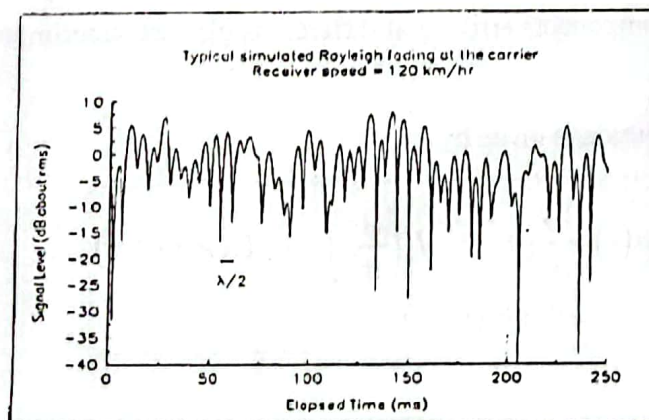


Rayleigh Fading Distribution

Rayleigh distribution is commonly used to describe the statistical time varying nature of the received envelope of a flat fading signal.

Rayleigh distribution also describe envelope of an individual multipath component.

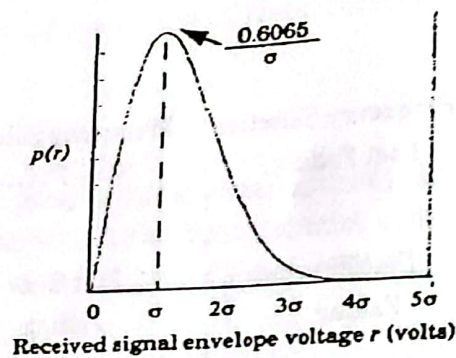
- Rayleigh distributed signal:



- ▶ The Rayleigh distribution has pdf

$$p(r) = \begin{cases} \frac{r}{\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) & (0 \leq r \leq \infty) \\ 0 & (r < 0) \end{cases}$$

σ = the rms value of the received voltage signal before envelope detection
 σ^2 = the time-average power of the received signal before envelope detection



- ▶ The probability that the envelope of the received signal does not exceed a specified value R is

$$P(R) = Pr(r \leq R) = \int_0^R p(r) dr = 1 - \exp\left(-\frac{R^2}{2\sigma^2}\right)$$

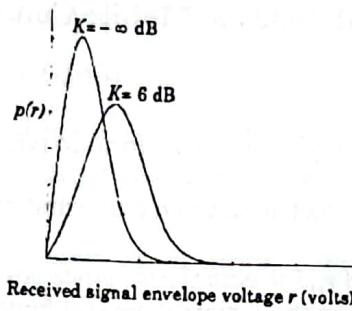
Rician Fading Distribution

When there is a dominant stationary (nonfading) signal component present, such as line-of-sight propagation path, the small-scale fading envelope distribution is Rician.

Random multipath components arriving at different angles are superimposed on a stationary dominant signal.

- ▶ The Rician distribution is given by

$$p(r) = \frac{r}{\sigma^2} e^{-\frac{(r^2 + A^2)}{2\sigma^2}} I_0\left(\frac{Ar}{\sigma^2}\right) \text{ for } (A \geq 0, r \geq 0) \\ = 0 \text{ for } (r < 0)$$



- The Rician distribution is described in terms of a parameter K

$$K = \frac{A^2}{2\sigma^2}$$

$$K(\text{dB}) = 10 \log \frac{A^2}{2\sigma^2}$$

- When $K = 0$ we have Rayleigh fading
- When $K = \infty$ we have no fading, channel has no multipath, only LOS component